

Assignment - 3

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⇒ Section - A

A1. The criteria of good estimator are unbiasedness, efficiency, consistency, sufficiency.

A2. Two methods of estimation are:
Method of Moments
Maximum likelihood estimation

A3. If MLE exists, it is the most efficient estimation in class of quick estimators.

A4. For Binomial distribution,

$$\Rightarrow \beta = \frac{\text{Sample mean } (\bar{x})}{\text{Theoretical Mean } (n)} \quad C(x) = np$$

$$\text{Var}(x) = np(1-p)$$

A5. $E(T_1) = q_1 + q_2$

$$E(T_2) = q_1 - q_2$$

$$E(T_1) + E(T_2) = q_1 + q_2 + q_1 - q_2$$

$$= 2q_1$$

$$\Rightarrow q_1 = \frac{T_1 + T_2}{2}$$

$$E(T_1) - E(T_2) = q_1 + q_2 - q_1 + q_2 = 2q_2$$

$$\Rightarrow q_2 = \frac{T_1 - T_2}{2}$$

A6. Circumstances :-

- ① Population is normally distributed
- ② Sample size is large
- ③ Variance is known.

A7. The interval $\left(\bar{y} - 1.96 \sqrt{\frac{s^2}{n}}, \bar{y} + 1.96 \sqrt{\frac{s^2}{n}} \right)$ is known as 95% confidence interval form, and the limits are $\bar{y} - 1.96 \sqrt{\frac{s^2}{n}}$ and $\bar{y} + 1.96 \sqrt{\frac{s^2}{n}}$.

A8. Under bayesian approach, the parameter of distribution is random variable.

A9. Bayes estimator is always a function of posterior statistics.

A10. (i) An estimator is unbiased if its expected value is equal to true value of the parameter it is estimating.

(ii) An estimator is said to be consistent if, as sample size increases, it converges in probability to the true parameter value.

(iii) An estimator is efficient if it has smallest possible variance among all unbiased estimators of the parameter.

(iv) An estimator is sufficient if it captures all the information about parameter that is present in the sample.

⇒ Section - B

B1. $X_1, X_2 \dots X_n \rightarrow$ random sample from $N(\mu, \sigma^2)$

$$f(x/\mu, \sigma^2) = \prod_{i=1}^n f(x_i/\mu, \sigma^2) = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2}$$

→ sufficient statistic for mean μ

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$$

→ sufficient estimator for variance σ^2

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2$$

B2. $X_1, X_2 \dots X_n \rightarrow$ uniform population $[0, q]$

$$f(x/q) = \begin{cases} 1/q, & 0 < x \leq q \\ 0, & \text{otherwise} \end{cases} = \prod_{i=1}^n f(x_i/q)$$

→ The max value of sample is

$$\hat{q} = \max(x_1, x_2, \dots, x_n)$$

B3.

$$N(\mu, \sigma^2)$$

(i) μ when σ^2 is known

→ Maximum likelihood estimator is $\bar{X}$.

(ii) σ^2 when μ is known.

$$\rightarrow \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

B4.

$$n = 100, \bar{X} = 0.048 \text{ inches}, \Delta = 0.01 \text{ inches},$$

$$\alpha = 1 - 0.99 = 0.01$$

$$Z_{0.005} \approx 2.576 \text{ and } SE = \frac{0.01}{10} = 0.001.$$

$$\Rightarrow \text{Margin of Error} = Z_{\alpha/2} \times SE.$$

$$= 0.001 \times 2.576 = 0.002576.$$

$$\Rightarrow \bar{X} + ME$$

$$\Rightarrow \text{Lower limit} : 0.048 + (-0.002576) = 0.045424.$$

$$\Rightarrow \text{upper limit} : 0.048 + 0.002576 = 0.050576$$

$$\Rightarrow (0.045424, 0.050576) \text{ inches.}$$

B5. For the poisson distribution, the general term $\frac{e^{-\mu} \mu^x}{x!}$

This implies that $\bar{x}$ is sufficient for μ .

$\Rightarrow$ Section - C

C1. $x_1 = 0.7$

$x_2 = 1.5$

$x_3 = 0.3$

$x_4 = 0.5$

$x_5 = 3.4$

$$\Rightarrow L(\theta) = \prod_{i=1}^5 \theta e^{-\theta} x_i = \theta^5 e^{-\theta \sum x_i}$$

Taking log on both sides,

$$\log L(\theta) = 5 \log \theta - \theta \sum_{i=1}^5 x_i$$

$$\frac{d}{d\theta} \log L(\theta) = \frac{5}{\theta} - \sum x_i = 0$$

$$\hat{\theta} = \frac{5}{\sum_{i=1}^5 x_i}$$

$$\Rightarrow \hat{\theta} = \frac{5}{0.7 + 1.5 + 0.3 + 0.5 + 3.4} = \frac{5}{6.4}$$

$$\Rightarrow \boxed{\hat{\theta} = 0.78125}$$

C3.

$$\text{Variance} = 44.1 \text{ inch}^2$$

$$\text{Confidence interval} = 95\%$$

$$n = 10$$

$$\bar{x} = (65 + 71 + 80 + 76 + 78 + 82 + 68 + 72 + 65 + 81) / 10.$$

$$\bar{x} = \frac{765}{10} = \underline{\underline{76.5}}$$

$$s^2 = 44.1$$

$$\Rightarrow s = 6.643$$

$$\Rightarrow SE = \frac{s}{\sqrt{n}} = \frac{6.643}{\sqrt{10}} = \underline{\underline{2.097}}$$

C4.

$$\alpha = 1 - 0.95 = 0.05$$

$$\alpha/2 = 0.025$$

$$Z_{0.025} = 1.96$$

$$ME = Z \times SE = 1.96 \times 2.097 = 4.112$$

$$\bar{x} + ME$$

$$\bar{x} = 76.5 \quad ME = 4.112$$

$$\Rightarrow \text{Lower limit} = 76.5 - 4.112 = 72.39$$

$$\Rightarrow \text{upper limit} = 76.5 + 4.112 = 80.612$$

$$\Rightarrow (72.39, 80.612) \text{ confidence interval.}$$

C4. To construct a 95% confidence interval,

$$\sigma^2 = 9 \text{ so } \sigma = 3$$

$$n = 25$$

$$\bar{x} = 30$$

$$\text{Tabulated value} = 1.96$$

$$\Rightarrow SE = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{25}} = 0.6$$

$$\Rightarrow \text{CI for } \mu = 30 \pm 1.96 \times 0.6 = 30 \pm 1.176 \Rightarrow (28.824, 31.176)$$

Transform into interval for $n(\theta) = 2\mu + 7$.

$$2(28.824) + 7 = 64.648, \quad 2(31.176) + 7 = 69.352$$

$\Rightarrow$ confidence interval for $n(\theta)$ is $(64.648, 69.352)$.

C5. The bayes estimator of p for a binomial random variable X with parameters n and p , and a uniform prior $f(p) = 1$, is derived from posterior distribution which follows a beta distribution $(x+1, n-x+1)$. The bayes estimator for p is the mean of this beta distribution given by

$$\Rightarrow \hat{p} = \frac{x+1}{n+2}$$

⇒ Section - D

D1.

Given: poisson distribution with PMF

$$P(x) = (e^{-m_1}) * m_1^x + e^{-m_2} * m_2^x / x!$$

$$(1) E(x) = m_1 + m_2$$

$$(2) E(x^2) = (m_1 + m_2) + (m_1^2 + m_2^2)$$

$$(2) \Rightarrow m_1 + m_2 = \mu_1 \text{ (sample mean)}$$

$$\Rightarrow (m_1 + m_2) + (m_1^2 + m_2^2) = \mu_2$$

Solving the system of equations:

$$(1) m_1 = \mu_1 + \sqrt{(\mu_2 - \mu_1^2)}$$

$$(2) m_2 = \mu_1 - \sqrt{(\mu_2 - \mu_1^2)}$$

D2.

Estimators T_1, T_2, T_3

Mean $\rightarrow \mu$

Variance $\rightarrow \sigma^2$

$$T_1 = x_1 + x_2 - x_3$$

$$T_2 = 2x_1 + 3x_3 - 4x_2$$

$$T_3 = \frac{1}{3} (x_1 + x_2 + x_3)$$

(i) Are T_1 and T_2 unbiased estimators?

$$E(T_1) = E(x_1 + x_2 - x_3)$$

$$= E(x_1) + E(x_2) - E(x_3)$$

$$= \mu + \mu + \mu$$

$$= \mu, \text{ unbiased.}$$

$$\begin{aligned}
 \Rightarrow E(T_2) &= E(2x_1 + 3x_3 - 4x_2) \\
 &= 2E(x_1) + 3E(x_3) - 4E(x_2) \\
 &= 2u + 3u - 5u \\
 &= u \rightarrow \text{unbiased.}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow T_3 &= \frac{1}{3} [x_1 + x_2 + x_3] \\
 &= \frac{1}{3} x_1 + \frac{1}{3} x_2 + \frac{1}{3} x_3 \\
 &= \frac{1}{3} E(x_1) + \frac{1}{3} E(x_2) + \frac{1}{3} E(x_3) \\
 &= \frac{1}{3} + \frac{1}{3} + \frac{1}{3}
 \end{aligned}$$

$$u = \frac{1}{3} (1 + 1 + 1)$$

$$3u = 1 (2 + 1)$$

$$3u = 2u + 1u$$

$$u = 1u$$

$$\Rightarrow \boxed{\lambda = 0}$$

DS $CI = \bar{x} \pm z \times \frac{s}{\sqrt{n}}$

$$\bar{x} = 2000 \quad n = 50$$

$$s = 540 \quad z = 1.645$$

$$\Rightarrow SE = \frac{s}{\sqrt{n}} = \frac{540}{\sqrt{50}} = \underline{\underline{76.4}}$$

$$\Rightarrow \text{Margin of error} = 2 \times SE$$

$$= 1.645 \times 76.4$$

$$= \underline{\underline{125.6}}$$

$$\Rightarrow CI = 2000 \pm 125.6$$

$$(2000 - 125.6, 2000 + 125.6) = (1874.4, 2125.6)$$

$\Rightarrow$ The 90% confidence interval for the true mean value of all sales invoice is approximately 1874.4 to 2125.6.

D4.

$$CI = \bar{x} \pm t \times \frac{s}{\sqrt{n}}$$

$$\bar{x} = \frac{11 + 37 + 25 + 62 + 51 + 21 + 18 + 43 + 32 + 20}{10}$$

$$= \frac{320}{10} = \underline{\underline{32}}$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{441 + 25 + 49 + 900 + 361 + 121 + 144}{9}$$

$$= \frac{2238}{9} \approx \underline{\underline{248.67}}$$

$$\Rightarrow s = \sqrt{248.67} \approx \underline{\underline{15.77}}$$

$$\Rightarrow SE = \frac{s}{\sqrt{n}} = \frac{15.77}{\sqrt{10}} = \underline{\underline{4.98}}$$

$$\Rightarrow ME = t \times SE = 1.833 \times 4.98 \approx \underline{\underline{9.12}}$$

$$CI = 32 \pm 9.12$$

$$\Rightarrow CI = (32 - 9.12, 32 + 9.12) = \boxed{(22.88, 41.22)}$$

D5. $CI = \bar{x} \pm z \times \frac{s}{\sqrt{n}}$

$$\bar{x} = 540$$

$$n = 81$$

$$s = 150$$

$$z = 1.96$$

$$\Rightarrow SE = \frac{s}{\sqrt{n}} = \frac{150}{\sqrt{81}} = \underline{\underline{16.67}}$$

$$\Rightarrow ME = z \times S.E = 1.96 \times 16.67 = \underline{\underline{32.67}}$$

$$\Rightarrow CI = 540 \pm 32.67$$

$$\Rightarrow CI = (540 - 32.67, 540 + 32.67) = \boxed{(507.33, 572.67)}$$

$\Rightarrow$ the 95% confidence interval for average amount per month by a family is approximately 507.33 to 572.67.